CHAPTER 10

how to delta hedge

The aim of this Chapter...

... is to show how classical calculus can be used to model situations in which you believe that there is arbitrage, and want to study risk and return.

In this Chapter...

- how to make money if your volatility forecast is more accurate than the market's
- different ways of delta hedging
- how much profit should you expect to make

10.1 INTRODUCTION

In this chapter I am going to boldly assert that there is money to be made from options, because options may be mispriced by the market. In simple terms, *there are arbitrage opportunities*. Shock, horror! I know that the whole of Chicago University has just thrown down this book in disgust. However, I hope the rest of you will enjoy the contents of this chapter for it puts into concrete mathematics some ideas that are most important, and frighteningly under-explained in the literature. This is the subject of how to delta hedge when your estimate of future actual volatility differs from that of the market as measured by the implied volatility.

As I hinted above, to some people, saying that actual volatility and implied volatility can be inconsistent with each other is a heresy, for it implies arbitrage and hence free money. Well, it's only as free as the model is accurate, that is, not at all. But even so, if there is such a difference (and vol arb hedge funds certainly think there is) then we can only get at that money by hedging, and if we have two estimates of volatility, which one goes into the famous Black–Scholes delta formula?

We'll see how you can hedge using a delta based on *either* actual volatility or on implied volatility. Neither is wrong, they just have different risk/return profiles.

If you do doubt that implied volatility and actual volatility can be in disagreement then take a look at Figure 10.1. This is simply a plot of the distributions of the logarithms of the VIX and of the rolling realized SPX volatility. The VIX is an implied volatility measure based on the SPX index and so you would expect it and the realized SPX volatility to bear some

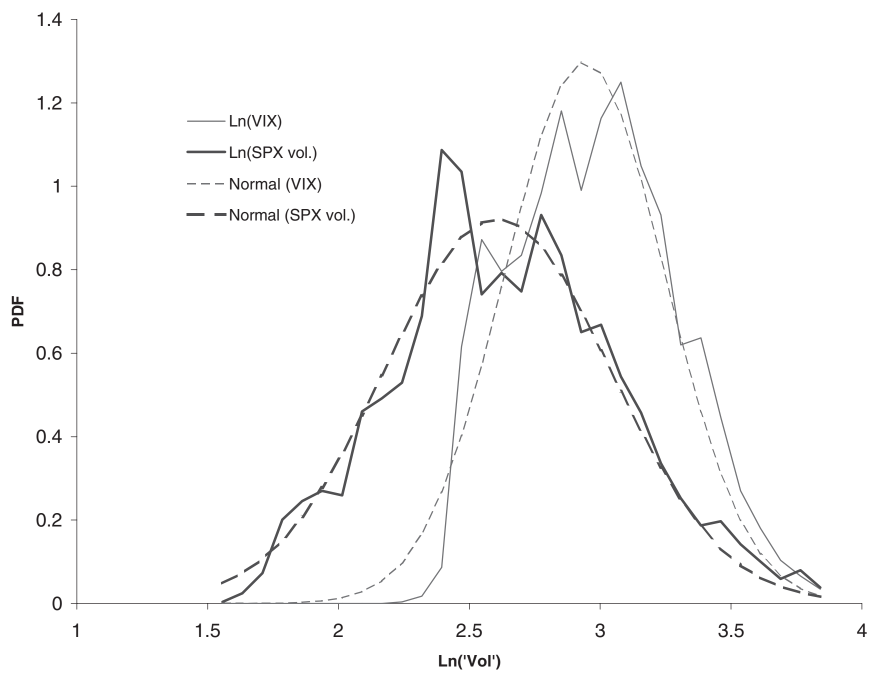


Figure 10.1 Distributions of the logarithms of the VIX and the rolling realized SPX volatility, and the Normal distributions for comparison.

resemblance. Not quite, as can be seen in the figure. The implied volatility VIX seems to be higher than the realized volatility. Both of these volatilities are approximately lognormally distributed (since their logarithms appear to be Normal), especially the realized volatility. The VIX distribution is somewhat truncated on the left. The mean of the realized volatility, about 15%, is significantly less than the mean of the VIX, about 20%, but its standard deviation is greater.

10.2 WHAT IF IMPLIED AND ACTUAL VOLATILITIES ARE DIFFERENT?

Actual volatility is the amount of 'noise' in the stock price, it is the coefficient of the Wiener process in the stock returns model, it is the amount of randomness that 'actually' transpires. Implied volatility is how the market is pricing the option currently. Since the market does not have perfect knowledge about the future these two numbers can and will be different.

Actual volatility being different from implied volatility is the heart of this chapter. Let's look at the simple case of exploiting such a difference by buying or selling options, but not delta hedging them.

Imagine that we have a forecast for volatility over the remaining life of an option, this volatility is forecast to be constant, and, crucially, our forecast turns out to be correct.

If you believe that actual volatility is higher than implied you might want to buy a straddle because there is then a good chance that the stock will move so far before expiry that you will get a payoff of more than the premium you paid, even after allowing for the time value of money. This is a very simple strategy, requiring no maintenance. There is one big problem with this however. It is risky. Sometimes you'll win, sometimes you'll lose. *Unless you can do this strategy many, many times you could end up losing a great deal.* Even if you are right about the actual volatility being large the stock might end up at the money, and you lose out. The relationship between actual volatility and the range over which an asset moves is a probabilistic one, there are no guarantees that a high volatility results in large moves.

If you buy an at-the-money straddle close to expiry the profit you expect to make from this strategy is approximately

$$\sqrt{\frac{2(T-t)}{\pi}}\,(\sigma-\tilde{\sigma})\,S.$$

The expression uses the close to expiry and ATM approximation we saw in Chapter 8. The notation is obvious; σ is the actual volatility, assumed constant, and $\tilde{\sigma}$ is the implied volatility. Note that this is an *expectation*. It is also a *real* expectation, however the real drift doesn't appear in this expression because it is an approximation valid only when close to expiration.

The standard deviation of the profit (the risk) is approximately

$$\sqrt{1-\frac{2}{\pi}}\quad \sigma S\sqrt{T-t}.$$

Observe how this depends on the actual volatility and not on the implied volatility. This standard deviation is of the same order of magnitude as the expected profit. That is a lot of risk. We can improve the risk-reward profile by delta hedging as we shall see next. The main purpose of writing down the above expressions is to show how they are linear in the two volatilities.

10.3 IMPLIED VERSUS ACTUAL, DELTA HEDGING BUT USING WHICH VOLATILITY?

Let's buy an underpriced option, or portfolio as above, but now, to improve risk and reward, we will delta hedge to expiry. This is a less risky strategy.

But which delta do you choose? Delta based on actual or implied volatility? This is one of those questions that people always ask, and one that no one seems to know the full answer to.

Scenario: Implied volatility for an option is 20%, but we believe that actual volatility is 30%.

Question: How can we make money if our forecast is correct?

Answer: Buy the option and delta hedge. But which delta do we use? We know that

$$\Delta = N(d_1)$$

where

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{s^2}{2}}\, ds$$

and

$$d_1 = \frac{\ln(S/E) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}.$$

We can all agree on S, E, $T-t$ and r (almost), but not on σ, should we use $\sigma = 0.2$ or 0.3, implied volatility or actual? In this example,

$$\sigma = \text{actual volatility, 30\%}$$

and

$$\tilde{\sigma} = \text{implied volatility, 20\%.}$$

Which of these goes into the d_1?

10.4 CASE 1: HEDGE WITH ACTUAL VOLATILITY, σ

By hedging with actual volatility we are replicating a short position in a *correctly priced* option. The payoffs for our long option and our short replicated option will exactly cancel. The profit we make will be exactly the difference in the Black–Scholes prices of an option with 30% volatility and one with 20% volatility. (Assuming that the Black–Scholes assumptions hold.) If $V(S, t; \sigma)$ is the Black–Scholes formula then the guaranteed profit is

$$V(S, t; \sigma) - V(S, t; \tilde{\sigma}).$$

But how is this guaranteed profit realized? Let's do the math on a mark-to-market basis.

In the following, superscript 'a' means actual and 'i' means implied, these can be applied to deltas and option values. For example, Δ^a is the delta using the actual volatility in the formula. V^i is the theoretical option value using the implied volatility in the formula. Note also that V, Δ, Γ and Θ are all simple, known, Black–Scholes formulæ.

The model is as usual

$$dS = \mu S\, dt + \sigma S\, dX.$$

Now, set up a portfolio by buying the option for V^i and hedge with Δ^a of the stock. The values of each of the components of our portfolio are shown in Table 10.1.

Leave this hedged portfolio overnight, and come back to it the next day. The new values are shown in Table 10.2. (I have included a continuous dividend yield in this.)

Therefore we have made, mark to market,

$$dV^i - \Delta^a\, dS - r(V^i - \Delta^a\, S)dt - \Delta^a DS\, dt.$$

Because the option would be correctly valued at V^a then we have

$$dV^a - \Delta^a\, dS - r(V^a - \Delta^a\, S)dt - \Delta^a DS\, dt = 0.$$

So we can write the mark-to-market profit over one time step as

$$\begin{aligned} &dV^i - dV^a + r(V^a - \Delta^a\, S)dt - r(V^i - \Delta^a\, S)dt \\ &= dV^i - dV^a - r(V^i - V^a)dt = e^{rt}\, d\left(e^{-rt}(V^i - V^a)\right). \end{aligned}$$

Table 10.1 Portfolio composition and values, today.

Component	Value
Option	V^i
Stock	$-\Delta^a\, S$
Cash	$-V^i + \Delta^a\, S$

Table 10.2 Portfolio composition and values, tomorrow.

Component	Value
Option	$V^i + dV^i$
Stock	$-\Delta^a\, S - \Delta^a\, dS$
Cash	$(-V^i + \Delta^a\, S)(1 + r\, dt) - \Delta^a DS\, dt$

That is the profit from time t to $t + dt$. The present value of this profit at time t_0 is

$$e^{-r(t-t_0)}e^{rt}\, d\left(e^{-rt}(V^i - V^a)\right) = e^{rt_0}\, d\left(e^{-rt}(V^i - V^a)\right).$$

So the total profit from t_0 to expiration is

$$e^{rt_0}\int_{t_0}^{T} d\left(e^{-rt}(V^i - V^a)\right) = V^a - V^i.$$

This confirms what I said earlier about the guaranteed total profit by expiration.

We can also write that one time step mark-to-market profit (using Itô's lemma) as

$$\begin{aligned}
&\Theta^i\, dt + \Delta^i\, dS + \tfrac{1}{2}\sigma^2 S^2\Gamma^i\, dt - \Delta^a\, dS - r(V^i - \Delta^a S)\, dt - \Delta^a DS\, dt \\
&\quad = \Theta^i\, dt + \mu S(\Delta^i - \Delta^a)\, dt + \tfrac{1}{2}\sigma^2 S^2\Gamma^i\, dt - r(V^i - V^a)\, dt + (\Delta^i - \Delta^a)\sigma S\, dX - \Delta^a DS\, dt \\
&\quad = (\Delta^i - \Delta^a)\sigma S\, dX + (\mu + D)S(\Delta^i - \Delta^a)\, dt + \tfrac{1}{2}\left(\sigma^2 - \tilde{\sigma}^2\right) S^2\Gamma^i\, dt
\end{aligned}$$

(using Black–Scholes with $\sigma = \tilde{\sigma}$)

$$= \tfrac{1}{2}\left(\sigma^2 - \tilde{\sigma}^2\right) S^2\Gamma^i\, dt + (\Delta^i - \Delta^a)\left((\mu - r + D)\, S\, dt + \sigma S\, dX\right).$$

The conclusion is that the final profit is guaranteed (the difference between the theoretical option values with the two volatilities) but how that is achieved is random, because of the dX term in the above. On a mark-to-market basis you could lose before you gain. Moreover, the mark-to-market profit depends on the real drift of the stock, μ. This is illustrated in Figure 10.2. The figure shows several realizations of the same delta-hedged

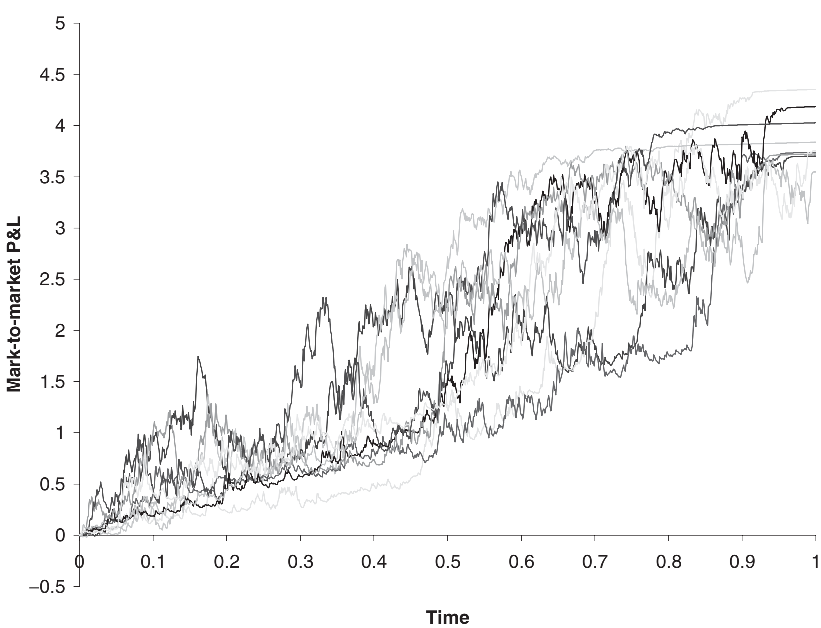


Figure 10.2 P&L for a delta-hedged option on a mark-to-market basis, hedged using actual volatility.

position. Note that the final P&L is not *exactly* the same in each case because of the effect of hedging discretely.

When S changes, so will V. But these changes do not cancel each other out. From a risk management point of view this is not ideal.

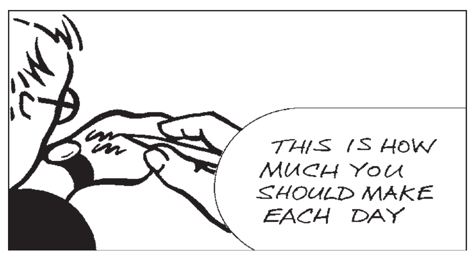
There is a simple analogy for this behavior. It is similar to owning a bond. For a bond there is a guaranteed outcome, but we may lose on a mark-to-market basis in the meantime.

10.5 CASE 2: HEDGE WITH IMPLIED VOLATILITY, $\tilde{\sigma}$

Compare and contrast now with the case of hedging using a delta based on implied volatility. By hedging with implied volatility we are balancing the random fluctuations in the mark-to-market option value with the fluctuations in the stock price. The evolution of the portfolio value is then 'deterministic' as we shall see.

Buy the option today, hedge using the implied delta, and put any cash in the bank earning r. The mark-to-market profit from today to tomorrow is

$$\begin{aligned} & dV^i - \Delta^i\, dS - r(V^i - \Delta^i S)dt - \Delta^i DS\, dt \\ &= \Theta^i\, dt + \tfrac{1}{2}\sigma^2 S^2 \Gamma^i\, dt - r(V^i - \Delta^i S)dt - \Delta^i DS\, dt \\ &= \tfrac{1}{2}\left(\sigma^2 - \tilde{\sigma}^2\right) S^2 \Gamma^i\, dt. \qquad (10.1) \end{aligned}$$



This is a far nicer way to make money. Observe how the daily profit is deterministic, there aren't any dX terms. From a risk management perspective this is much better behaved. There is another, rather wonderful, advantage of hedging using implied volatility . . . we don't even need to know what actual volatility is. And to make a profit all we need to know is that actual is always going to be greater than implied (if we are buying) or always less (if we are selling). This takes some of the pressure off forecasting volatility accurately in the first place.

Add up the present value of all of these profits to get a total profit of

$$\tfrac{1}{2}\left(\sigma^2 - \tilde{\sigma}^2\right) \int_{t_0}^{T} e^{-r(t-t_0)} S^2 \Gamma^i\, dt.$$

This is always positive, but highly path dependent. Being path dependent it will depend on the drift μ. If we start off at the money and the drift is very large (positive or negative) we will find ourselves quickly moving into territory where gamma and hence (10.1) is small, so that there will be not much profit to be made. The best that could happen would be for the stock to end up close to the strike at expiration, this would maximize the total profit. This path dependency is shown in Figure 10.3. The figure shows several realizations of the same delta-hedged position. Note that the lines are not perfectly smooth, again because of the effect of hedging discretely.

The simple analogy is now just putting money in the bank. The P&L is always increasing in value but the end result is random.

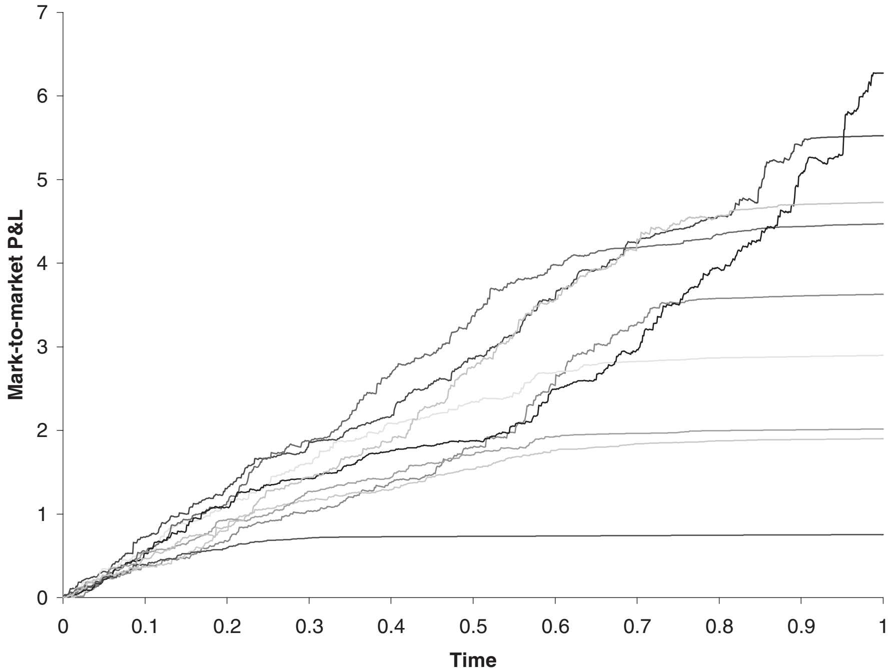


Figure 10.3 P&L for a delta-hedged option on a mark-to-market basis, hedged using implied volatility.

Peter Carr (2005) and Henrard (2001) show that if you hedge using a delta based on a volatility σ_h then the PV of the total profit is given by

$$V(S,t;\sigma_h) - V(S,t;\tilde{\sigma}) + \frac{1}{2}\left(\sigma^2 - \sigma_h^2\right)\int_{t_0}^{T} e^{-r(t-t_0)} S^2 \Gamma^h \, dt, \tag{10.2}$$

where the superscript on the gamma means that it uses the Black–Scholes formula with a volatility of σ_h.

10.5.1 The expected profit after hedging using implied volatility

When you hedge using delta based on implied volatility the profit each 'day' is deterministic but the present value of total profit by expiration is path dependent, and given by

$$\frac{1}{2}(\sigma^2 - \tilde{\sigma}^2)\int_{t_0}^{T} e^{-r(s-t_0)} S^2 \Gamma^i \, ds.$$

The details of the analysis of this expression are found in *PWOQF2*. Anyway, after some manipulations we end up with the expected profit initially being the single integral

$$\frac{Ee^{-r(T-t_0)}(\sigma^2-\tilde{\sigma}^2)}{2\sqrt{2\pi}}\int_{t_0}^{T}\frac{1}{\sqrt{\sigma^2(s-t_0)+\tilde{\sigma}^2(T-s)}}$$

$$\exp\left(-\frac{\left(\log(S/E)+\left(\mu-\frac{1}{2}\sigma^2\right)(s-t_0)+\left(r-D-\frac{1}{2}\tilde{\sigma}^2\right)(T-s)\right)^2}{2(\sigma^2(s-t_0)+\tilde{\sigma}^2(T-s))}\right)ds.$$

A bit complicated, but implemented on the CD

Results are shown in Figures 10.4–10.6.

In Figure 10.4 is shown the expected profit versus the growth rate μ. Parameters are $S=100$, $\sigma=0.4$, $r=0.05$, $D=0$, $E=110$, $T=1$, $\tilde{\sigma}=0.2$. Observe that the expected profit has a maximum. This will be at the growth rate that ensures, roughly speaking, that the stock ends up close to at the money at expiration, where gamma is largest. In the figure is also shown the profit to be made when hedging with actual volatility. For most realistic parameters regimes the maximum expected profit hedging with implied is similar to the guaranteed profit hedging with actual.

In Figure 10.5 is shown expected profit versus E and μ. You can see how the higher the growth rate the larger the strike price at the maximum. The contour map is shown in Figure 10.6.

10.5.2 The variance of profit after hedging using implied volatility

Once we have calculated the expected profit from hedging using implied volatility we can calculate the variance in the final profit. Again, all details may be found in *PWOQF2*.

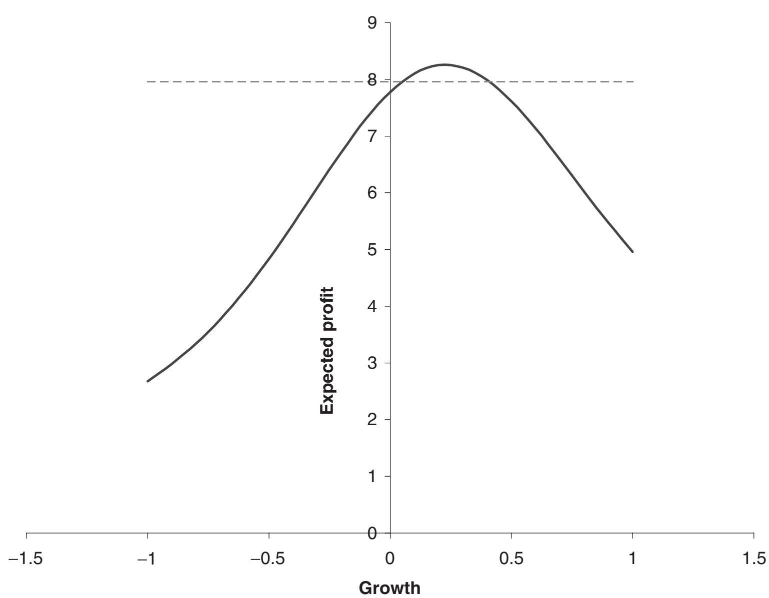


Figure 10.4 Expected profit, hedging using implied volatility, versus growth rate μ; $S=100$, $\sigma=0.4$, $r=0.05$, $D=0$, $E=110$, $T=1$, $\tilde{\sigma}=0.2$. The dashed line is the profit to be made when hedging with actual volatility.

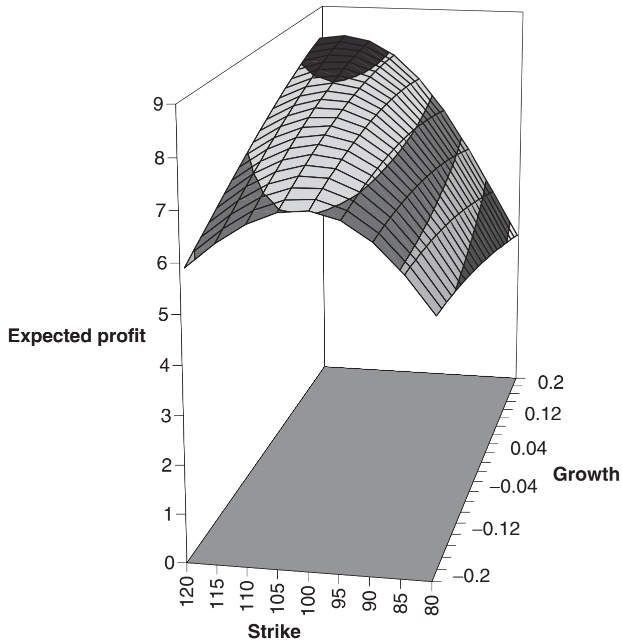


Figure 10.5 Expected profit, hedging using implied volatility, versus growth rate μ and strike E; $S = 100$, $\sigma = 0.4$, $r = 0.05$, $D = 0$, $T = 1$, $\tilde{\sigma} = 0.2$.

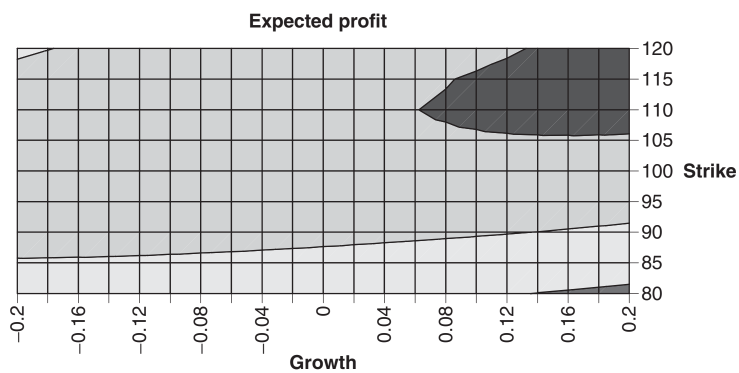


Figure 10.6 Contour map of expected profit, hedging using implied volatility, versus growth rate μ and strike E; $S = 100$, $\sigma = 0.4$, $r = 0.05$, $D = 0$, $T = 1$, $\tilde{\sigma} = 0.2$.

The initial variance is $G(S_0, t_0) - F(S_0, t_0)^2$, where

$$G(S_0, t_0) = \frac{E^2(\sigma^2 - \tilde{\sigma}^2)^2 e^{-2r(T-t_0)}}{4\pi\sigma\tilde{\sigma}} \int_{t_0}^{T}\int_{s}^{T}$$

$$\times \frac{e^{p(u,s;S_0,t_0)}}{\sqrt{s-t_0}\sqrt{T-s}\sqrt{\sigma^2(u-s)+\tilde{\sigma}^2(T-u)}\sqrt{\frac{1}{\sigma^2(s-t_0)}+\frac{1}{\tilde{\sigma}^2(T-s)}+\frac{1}{\sigma^2(u-s)+\tilde{\sigma}^2(T-u)}}}\, du\, ds \quad (10.3)$$

where

$$p(u,s;S_0,t_0) = -\frac{1}{2}\frac{(x+\alpha(T-s))^2}{\tilde{\sigma}^2(T-s)} - \frac{1}{2}\frac{(x+\alpha(T-u))^2}{\sigma^2(u-s)+\tilde{\sigma}^2(T-u)}$$

$$+\frac{1}{2}\frac{\left(\dfrac{x+\alpha(T-s)}{\tilde{\sigma}^2(T-s)}+\dfrac{x+\alpha(T-u)}{\sigma^2(u-s)+\tilde{\sigma}^2(T-u)}\right)^2}{\dfrac{1}{\sigma^2(s-t_0)}+\dfrac{1}{\tilde{\sigma}^2(T-s)}+\dfrac{1}{\sigma^2(u-s)+\tilde{\sigma}^2(T-u)}}$$

Even more complicated, but still implemented on the CD

and

$$x = \ln(S_0/E) + \left(\mu - \tfrac{1}{2}\sigma^2\right)(T-t_0), \quad \text{and} \quad \alpha = \mu - \tfrac{1}{2}\sigma^2 - r + D + \tfrac{1}{2}\tilde{\sigma}^2.$$

In Figure 10.7 is shown the standard deviation of profit versus growth rate, $S = 100$, $\sigma = 0.4$, $r = 0.05$, $D = 0$, $E = 110$, $T = 1$, $\tilde{\sigma} = 0.2$. Figure 10.8 shows the standard deviation of profit versus strike, $S = 100$, $\sigma = 0.4$, $r = 0.05$, $D = 0$, $\mu = 0.1$, $T = 1$, $\tilde{\sigma} = 0.2$.

Note that in these plots the expectations and standard deviations have not been scaled with the cost of the options.

10.6 HEDGING WITH DIFFERENT VOLATILITIES

We will briefly examine hedging using volatilities other than actual or implied, using the general expression for profit given by (10.2).

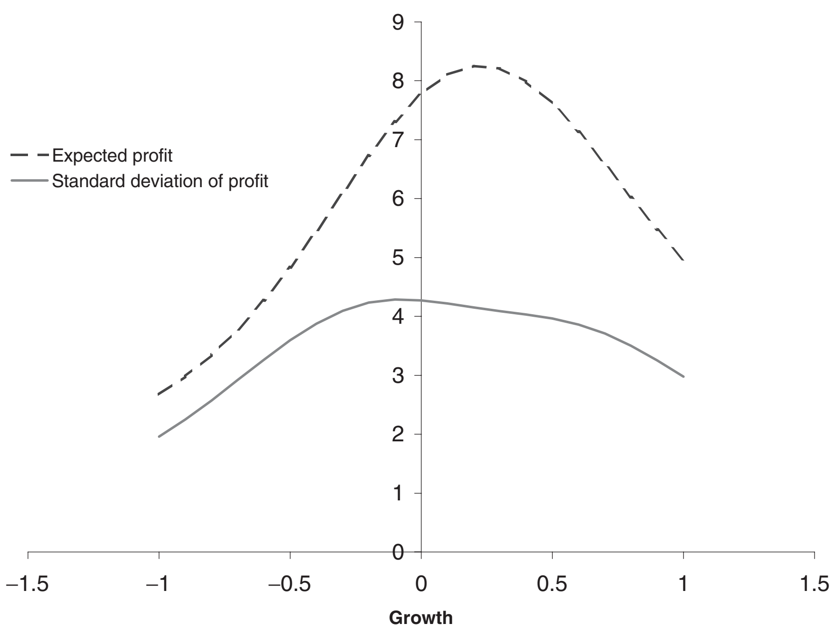


Figure 10.7 Standard deviation of profit, hedging using implied volatility, versus growth rate μ; $S = 100$, $\sigma = 0.4$, $r = 0.05$, $D = 0$, $E = 110$, $T = 1$, $\tilde{\sigma} = 0.2$. (The expected profit is also shown.)

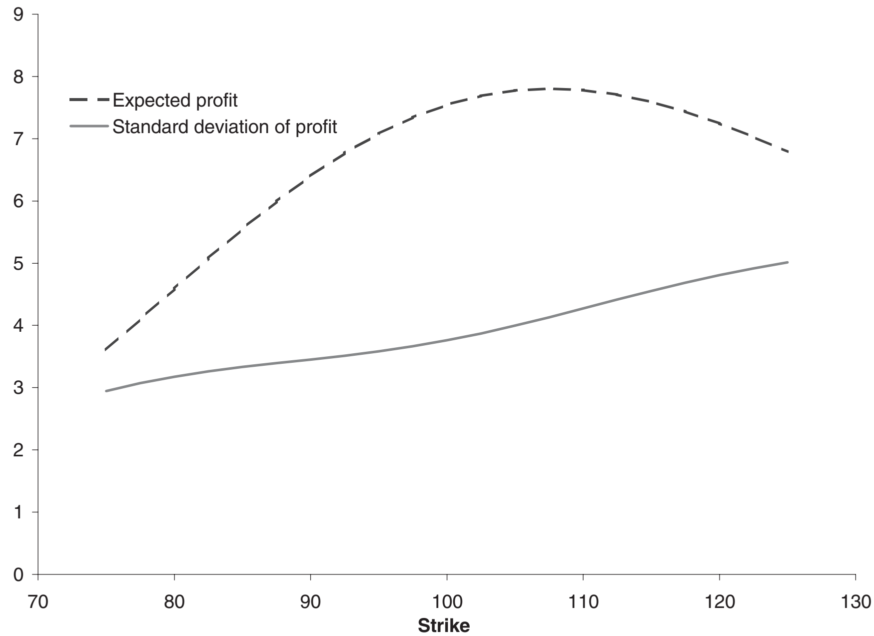


Figure 10.8 Standard deviation of profit, hedging using implied volatility, versus strike E; $S = 100$, $\sigma = 0.4$, $r = 0.05$, $D = 0$, $\mu = 0$, $T = 1$, $\tilde{\sigma} = 0.2$. (The expected profit is also shown.)

The expressions for the expected profit and standard deviations now must allow for the $V(S, t; \sigma_h) - V(S, t; \tilde{\sigma})$, since the integral of gamma term can be treated as before if one replaces $\tilde{\sigma}$ with σ_h in this term. Results are presented in the next sections.

10.6.1 Actual volatility = Implied volatility

For the first example let's look at hedging a long position in a correctly priced option, so that $\sigma = \tilde{\sigma}$. We will hedge using different volatilities, σ^h. Results are shown in Figure 10.9. The figure shows the expected profit and standard deviation of profit when hedging with various volatilities. The chart also shows minimum and maximum profit. Parameters are $E = 100$, $S = 100$, $\mu = 0$, $\sigma = 0.2$, $r = 0.1$, $D = 0$, $T = 1$, and $\tilde{\sigma} = 0.2$.

With these parameters the expected profit is small as a fraction of the market price of the option (\$13.3) regardless of the hedging volatility. The standard deviation of profit is zero when the option is hedged at the actual volatility. The upside, the maximum profit is much greater than the downside. Crucially all of the curves have zero value at the actual/implied volatility.

10.6.2 Actual volatility > Implied volatility

In Figure 10.10 is shown the expected profit and standard deviation of profit when hedging with various volatilities when actual volatility is greater than implied. The chart again also shows minimum and maximum profit. Parameters are $E = 100$, $S = 100$, $\mu = 0$, $\sigma = 0.4$, $r = 0.1$, $D = 0$, $T = 1$, and $\tilde{\sigma} = 0.2$. Note that it is possible to lose money if you hedge at

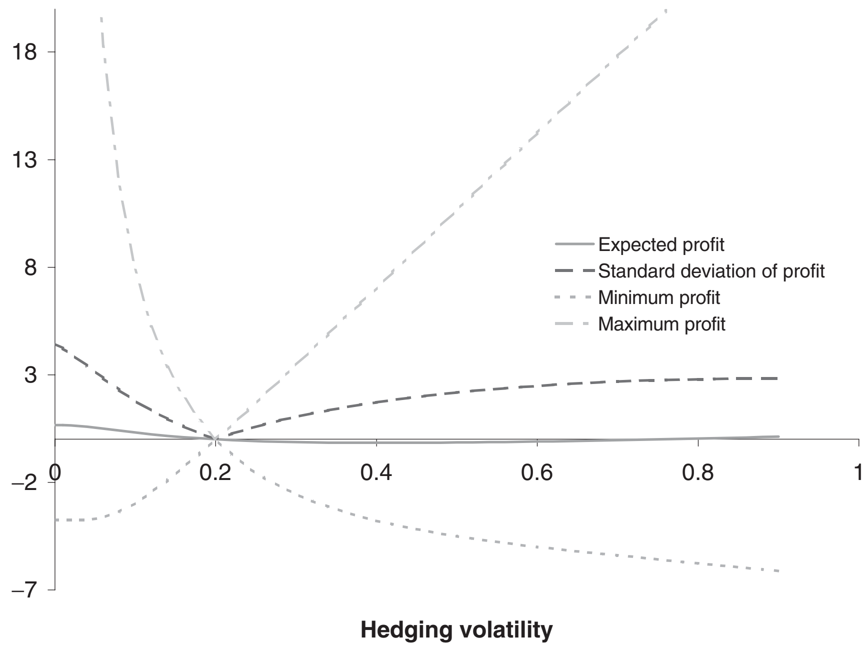


Figure 10.9 Expected profit, standard deviation of profit, minimum and maximum, hedging with various volatilities. $E = 100$, $S = 100$, $\mu = 0$, $\sigma = 0.2$, $r = 0.1$, $D = 0$, $T = 1$, $\tilde{\sigma} = 0.2$.

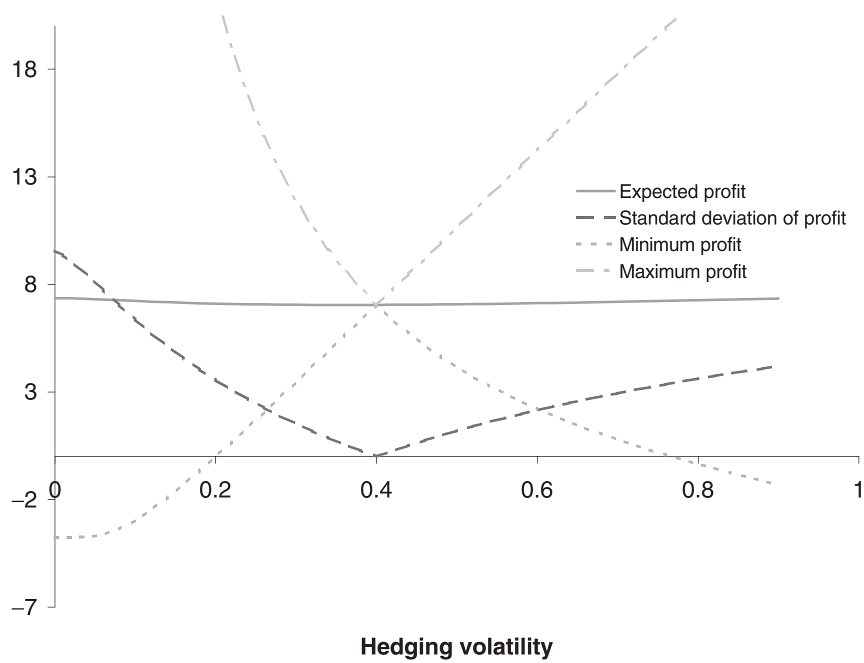


Figure 10.10 Expected profit, standard deviation of profit, minimum and maximum, hedging with various volatilities. $E = 100$, $S = 100$, $\mu = 0$, $\sigma = 0.4$, $r = 0.1$, $D = 0$, $T = 1$, $\tilde{\sigma} = 0.2$.

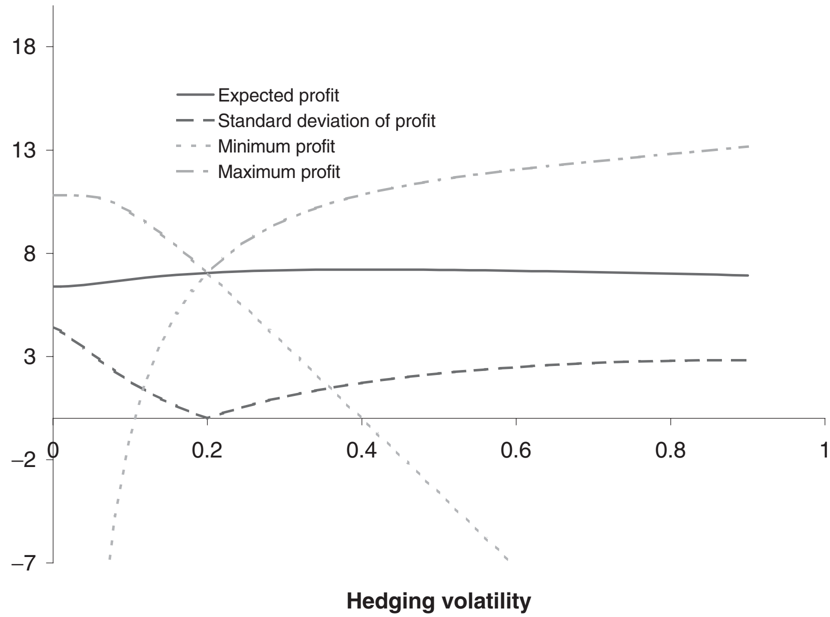


Figure 10.11 Expected profit, standard deviation of profit, minimum and maximum, hedging with various volatilities. $E = 100$, $S = 100$, $\mu = 0$, $\sigma = 0.2$, $r = 0.1$, $D = 0$, $T = 1$, $\tilde{\sigma} = 0.4$.

below implied, but hedging with a higher volatility you will not be able to lose until hedging with a volatility of approximately 75%. The expected profit is again insensitive to hedging volatility.

10.6.3 Actual volatility < Implied volatility

In Figure 10.11 is shown properties of the profit when hedging with various volatilities when actual volatility is less than implied. We are now selling the option and delta hedging it. Parameters are $E = 100$, $S = 100$, $\mu = 0$, $\sigma = 0.4$, $r = 0.1$, $D = 0$, $T = 1$, and $\tilde{\sigma} = 0.2$. Now it is possible to lose money if you hedge at above implied, but hedging with a lower volatility you will not be able to lose until hedging with a volatility of approximately 10%. The expected profit is again insensitive to hedging volatility. The downside is now more dramatic than the upside.

10.7 PROS AND CONS OF HEDGING WITH EACH VOLATILITY

Given that we seem to have a choice in how to delta hedge it is instructive to summarize the advantages and disadvantages of the possibilities.

10.7.1 Hedging with actual volatility

Pros: The main advantage of hedging with actual volatility is that you know exactly what profit you will get at expiration. So in a classical risk/reward sense this seems to be the best choice, given that the expected profit can often be insensitive to which volatility you choose to hedge with whereas the standard deviation is always going to be positive away from hedging with actual volatility.

Cons: The P&L fluctuations during the life of the option can be daunting, and so less appealing from a 'local' as opposed to 'global' risk management perspective. Also, you are unlikely to be totally confident in your volatility forecast, the number you are putting into your delta formula. However, you can interpret the previous two figures in terms of what happens if you intend to hedge with actual but don't quite get it right. You can see from those that you do have quite a lot of leeway before you risk losing money.

10.7.2 Hedging with implied volatility

Pros: There are three main advantages to hedging with implied volatility. The first is that there are no local fluctuations in P&L, you are continually making a profit. The second advantage is that you only need to be on the right side of the trade to profit. Buy when actual is going to be higher than implied and sell if lower. Finally, the number that goes into the delta is implied volatility, and therefore easy to observe.

Cons: You don't know how much money you will make, only that it is positive.

10.7.3 Hedging with another volatility

You can obviously balance the pros and cons of hedging with actual and implied by hedging with another volatility altogether. See Dupire (2005) for work in this area.

In practice which volatility one uses is often determined by whether one is constrained to mark to market or mark to model. If one is able to mark to model then one is not necessarily concerned with the day-to-day fluctuations in the mark-to-market profit and loss and so it is natural to hedge using actual volatility. This is usually not far from optimal in the sense of possible expected total profit, and it has no standard deviation of final profit. However, it is common to have to report profit and loss based on market values. This constraint may be imposed by a risk management department, by prime brokers, or by investors who may monitor the mark-to-market profit on a regular basis. In this case it is more usual to hedge based on implied volatility to avoid the daily fluctuations in the profit and loss.

For the remainder of this chapter we will only consider the case of hedging using a delta based on implied volatility.

10.8 **PORTFOLIOS WHEN HEDGING WITH IMPLIED VOLATILITY**

A natural extension to the above analysis is to look at portfolios of options, options with different strikes and expirations. Since only an option's gamma matters when we are hedging using implied volatility, calls and puts are effectively the same since they have the same gamma.

The profit from a portfolio is now

$$\tfrac{1}{2}\sum_k q_k\left(\sigma^2-\tilde{\sigma}_k^2\right)\int_{t_0}^{T_k} e^{-r(s-t_0)}S^2\Gamma_k^i\,ds,$$

where k is the index for an option, and q_k is the quantity of that option.

10.8.1 Expectation

The solution for the present value of the expected profit ($t=t_0$, $S=S_0$, $I=0$) is simply the sum of individual profits for each option,

$$F(S_0,t_0)=\sum_k q_k\frac{E_k e^{-r(T_k-t_0)}(\sigma^2-\tilde{\sigma}_k^2)}{2\sqrt{2\pi}}\int_{t_0}^{T_k}\frac{1}{\sqrt{\sigma^2(s-t_0)+\tilde{\sigma}_k^2(T_k-s)}}$$
$$\times\exp\left(-\frac{\left(\ln(S_0/E_k)+\left(\mu-\frac{1}{2}\sigma^2\right)(s-t_0)+\left(r-D-\frac{1}{2}\tilde{\sigma}_k^2\right)(T_k-s)\right)^2}{2(\sigma^2(s-t_0)+\tilde{\sigma}_k^2(T_k-s))}\right)\,ds.$$

10.8.2 Variance

The variance is more complicated, obviously, because of the correlation between all of the options in the portfolio. Nevertheless, we can find an expression for the initial variance as $G(S_0,t_0)-F(S_0,t_0)^2$ where

$$G(S_0,t_0)=\sum_j\sum_k q_j q_k G_{jk}(S_0,t_0)$$

where

$$G_{jk}(S_0,t_0)=\frac{E_jE_k(\sigma^2-\tilde{\sigma}_j^2)(\sigma^2-\tilde{\sigma}_k^2)e^{-r(T_j-t_0)-r(T_k-t_0)}}{4\pi\,\sigma\,\tilde{\sigma}_k}\int_{t_0}^{\min(T_j,T_k)}\int_s^{T_j}$$
$$\times\frac{e^{p(u,s;S_0,t_0)}}{\sqrt{s-t_0}\sqrt{T_k-s}\sqrt{\sigma^2(u-s)+\tilde{\sigma}_j^2(T_j-u)}\sqrt{\frac{1}{\sigma^2(s-t_0)}+\frac{1}{\tilde{\sigma}_k^2(T_k-s)}+\frac{1}{\sigma^2(u-s)+\tilde{\sigma}_j^2(T_j-u)}}}\,du\,ds \tag{10.4}$$

where

$$p(u,s;S_0,t_0)=-\frac{1}{2}\frac{(\ln(S_0/E_k)+\overline{\mu}(s-t_0)+\overline{r}_k(T_k-s))^2}{\tilde{\sigma}_k^2(T_k-s)}$$
$$-\frac{1}{2}\frac{(\ln(S_0/E_j)+\overline{\mu}(u-t_0)+\overline{r}_j(T_j-u))^2}{\sigma^2(u-s)+\tilde{\sigma}_j^2(T_j-u)}$$
$$+\frac{1}{2}\frac{\left(\frac{\ln(S_0/E_k)+\overline{\mu}(s-t_0)+\overline{r}_k(T_k-s)}{\tilde{\sigma}_k^2(T_k-s)}+\frac{\ln(S_0/E_j)+\overline{\mu}(u-t_0)+\overline{r}_j(T_j-u)}{\sigma^2(u-s)+\tilde{\sigma}_j^2(T_j-u)}\right)^2}{\frac{1}{\sigma^2(s-t_0)}+\frac{1}{\tilde{\sigma}_k^2(T_k-s)}+\frac{1}{\sigma^2(u-s)+\tilde{\sigma}_j^2(T_j-u)}}$$

and

$$\overline{\mu} = \mu - \tfrac{1}{2}\sigma^2, \quad \overline{r}_j = r - D - \tfrac{1}{2}\tilde{\sigma}_j^2 \quad \text{and} \quad \overline{r}_k = r - D - \tfrac{1}{2}\tilde{\sigma}_k^2.$$

10.8.3 Portfolio optimization possibilities

There is clearly plenty of scope for using the above formulae in portfolio optimization problems. Here I give one example.

The stock is currently at 100. The growth rate is zero, actual volatility is 20%, zero dividend yield and the interest rate is 5%. Table 10.3 shows the available options, and associated parameters. Observe the negative skew. The out-of-the-money puts are overvalued and the out-of-the-money calls are undervalued. (The ‘Profit total expected’ row assumes that we buy a single one of that option.)

Using the above formulae we can find the portfolio that maximizes or minimizes target quantities (expected profit, standard deviation, ratio of profit to standard deviation). Let us consider the simple case of maximizing the expected return, while constraining the standard deviation to be one. This is a very natural strategy when trying to make a profit from volatility arbitrage while meeting constraints imposed by regulators, brokers, investors, etc. The result is given in Table 10.4.

The payoff function (with its initial delta hedge) is shown in Figure 10.12. This optimization has effectively found an ideal risk reversal trade. This portfolio would cost −\$0.46 to set up, i.e. it would bring in premium. The expected profit is \$6.83.

10.9 HOW DOES IMPLIED VOLATILITY BEHAVE?

Now is the natural time to talk a little bit about how implied volatility behaves in practice.

As the stock price goes up and down randomly we often see that the implied volatility of each option will also vary. This may or may not be consistent with certain models, and

Table 10.3 Available options.

	A	B	C	D	E
Type	Put	Put	Call	Call	Call
Strike	80	90	100	110	120
Expiration	1	1	1	1	1
Volatility, implied	0.250	0.225	0.200	0.175	0.150
Option price, market	1.511	3.012	10.451	5.054	1.660
Option value, theory	0.687	2.310	10.451	6.040	3.247
Profit total expected	−0.933	−0.752	0.000	0.936	1.410

Table 10.4 An optimal portfolio.

	A	B	C	D	E
Type	Put	Put	Call	Call	Call
Strike	80	90	100	110	120
Quantity	−2.10	−2.25	0	1.46	1.28

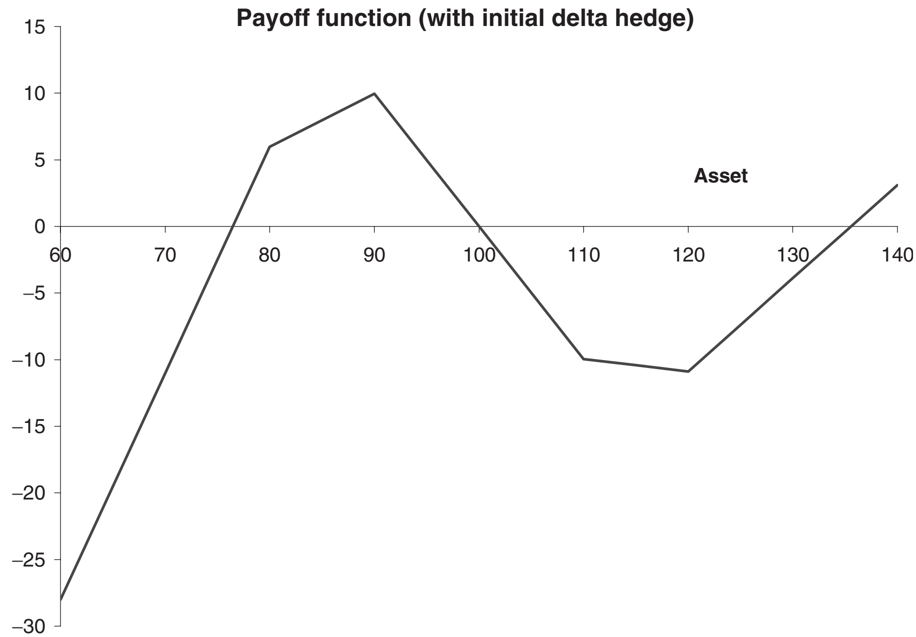


Figure 10.12 Payoff with initial delta hedge for optimal portfolio; $S = 100$, $\mu = 0$, $\sigma = 0.2$, $r = 0.05$, $D = 0$, $T = 1$. See text for additional parameters and information.

may or may not be consistent with no arbitrage. But more importantly, what does it mean for making money if we think that the market is wrong? Below are a couple of 'models' for how implied volatility might change as the market moves.

10.9.1 Sticky strike

In this model implied volatility remains constant for each option (i.e. each strike and expiration). Effectively each option inhabits its own little Black–Scholes world of constant volatility. This behavior seems to be most common in the equity markets. As far as making a profit if the implied volatility is different from actual volatility then the first half of this chapter is clearly very relevant.

10.9.2 Sticky delta

Since the delta of an option is a function of its moneyness, S/E, the sticky delta behavior could also be called the sticky moneyness rule. This behavior is commonly seen in the FX markets, possibly because there it is usual to quote option prices/volatilities for options with specific deltas rather than specific strike. (There is, of course, a one-to-one correspondence for vanillas.)

In this model we have

$$\tilde{\sigma} = g(S/E, t).$$

Therefore

$$d\tilde{\sigma} = \left(\frac{\partial g}{\partial t} + \mu \frac{S}{E} \frac{\partial g}{\partial \xi} + \tfrac{1}{2}\sigma^2 \frac{S^2}{E^2} \frac{\partial^2 g}{\partial \xi^2} \right) dt + \sigma \frac{S}{E} \frac{\partial g}{\partial \xi} \, dX_1,$$

where $\xi = S/E$. The most important point about this expression is that it is perfectly correlated with dS, $\rho = 1$, so that perfect hedging (in the mark-to-market sense) is possible.

A variation on this theme is to have implied volatility at different strikes being proportional to the ATM volatility and a function of the moneyness, such as

$$\tilde{\sigma} = \sigma_{ATM} \, g\left(\frac{\log(S/E)}{\sqrt{T-t}} \right).$$

See Natenberg (1994) for details of this. Of course, this then requires a model for the behavior of the ATM volatility.

10.9.3 Time-periodic behavior

Just to make matters even more interesting, there appears to be a day-of-the-week effect in implied volatility. The next few figures show how the VIX volatility index (a measure of the implied volatility of the ATM SPX adjusted to always have an expiration of 30 days)

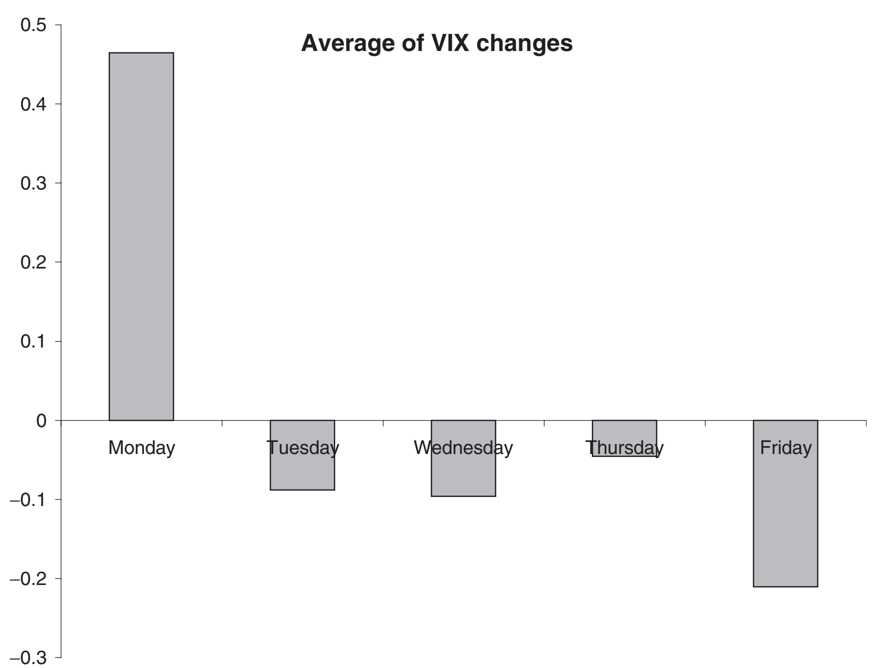


Figure 10.13 Average change in level of VIX versus day of week.

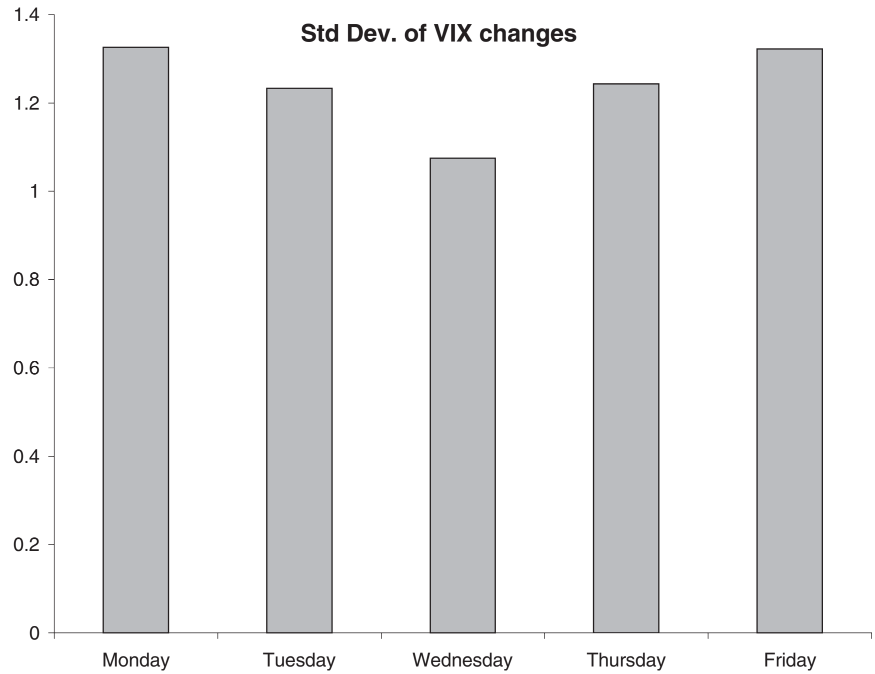


Figure 10.14 Standard deviation of change in level of VIX versus day of week.

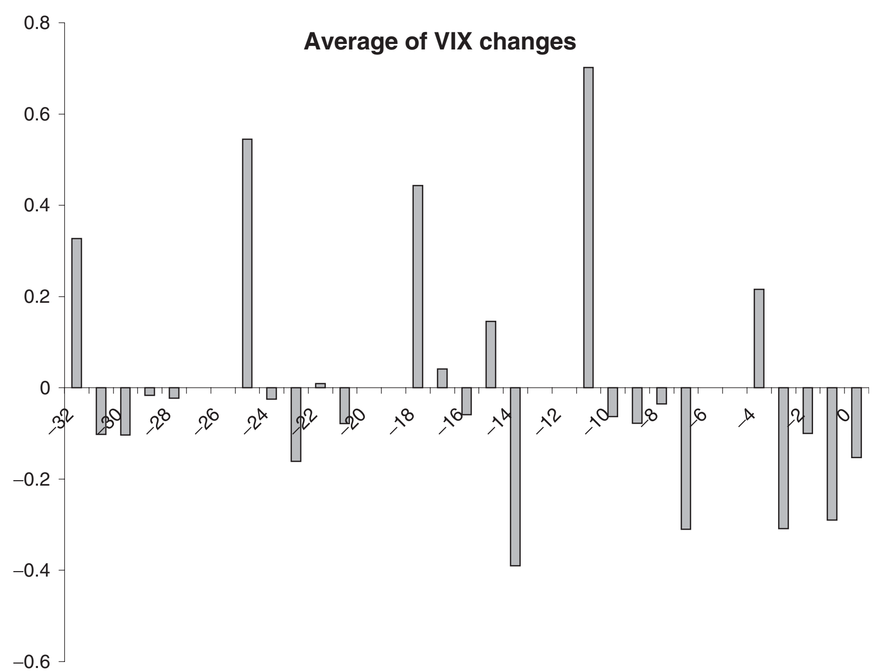


Figure 10.15 Average change in level of VIX versus days before next expiration.

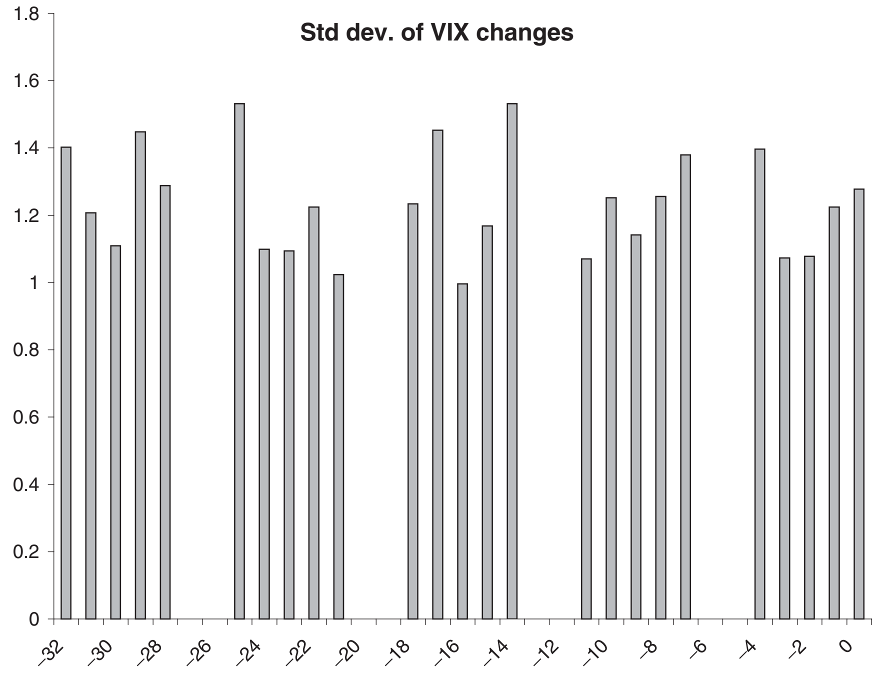


Figure 10.16 Standard deviation of change in level of VIX versus days before next expiration.

changes with day of the week and number of days to next expiration. Both average changes and standard deviation are shown.[1]

10.10 SUMMARY

In this chapter we have seen some hints at how we can start to move away from the Black–Scholes world, and perhaps even profit from options.

FURTHER READING

- See Derman (1999) for a description of sticky strike and delta, and other volatility regimes.
- Natenberg's book (Natenberg, 1994) is still the classic reference for volatility trading.
- See Carr's (Carr, 2005) excellent FAQs paper for further insight into which volatility to use for hedging. Also Henrard (2001), who examined the role of the real drift rate.
- Ahmad & Wilmott (2005) delve even deeper into the subject of hedging with different volatilities.

[1] Of course, some of this is no doubt related to the role of weekends in the calculation of volatility. There is a lot of 'potential' for volatility over weekends in the sense that there is plenty of news that comes out that will impact on market prices when markets open on the Monday.

EXERCISES

1. Take the How to Hedge spreadsheet on the CD and rewrite using VB, C++, or other code. Now modify the code to do the following.
   (a) Allow for arbitrary fixed period between rehedges. Observe how the hedging error varies with this period.
   (b) Incorporate bid-offer spread on each transaction in the underlying.
   (c) As above but now for the delta hedging of an entire portfolio of vanilla options of varying type, strikes and expiration.

   Write the code so that you can perform many thousands of simulations, and output statistical properties of the hedging error.